

The Internet Effect and Labor's New Era

The Impact of Internet-Driven Productivity Acceleration on Labor's Marginal Contribution to GDP Growth and the Shift in Labor-Productivity Elasticity through the Solow-Swan Model



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Abstract

This econometric research paper delves into the transformative impact of the internet revolution on labor's contribution to GDP growth, with a particular focus on the post-1995 era. The primary objective is to ascertain whether the internet-driven acceleration in productivity has led to a structural decline in labor's marginal contribution to GDP or, conversely, if it has enhanced labor's complementarity with technological progress. By examining the shift in labor-productivity elasticity, the study seeks to understand if the impact of productivity on labor's contribution to GDP has increased due to the internet. Utilizing a range of statistical tools, including multiple linear OLS estimation, interaction terms, Chow Test, and t-tests, the paper analyzes the relationship between internet penetration, technological adoption, and labor productivity. The findings aim to provide a comprehensive understanding of how the internet has reshaped labor dynamics and productivity, offering valuable insights for policy-making and future economic research.

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1 Introduction

The Solow-Swan model, a cornerstone of growth theory developed in 1956, revolutionized our understanding of economic growth by emphasizing the crucial role of technological progress. This neoclassical framework demonstrated that beyond traditional inputs of capital and labor, technological advancement serves as the primary driver of sustained economic expansion. While the model's implications have been extensively studied in various technological contexts, perhaps no technological shift has provided a more compelling testing ground for its predictions than the digital revolution of the late 20th century.

The advent of the internet has marked one of the most profound technological revolutions in modern economic history, fundamentally altering production processes, business operations, and labor dynamics. Since the mid-1990s, rapid advancements in digital connectivity, automation, and information exchange have redefined the relationship between productivity and labor, raising important questions about the evolving nature of economic growth. This study investigates whether the internet-driven acceleration in productivity has led to a structural decline in labor's marginal contribution to GDP growth or, conversely, if it has reinforced labor's complementarity with technological progress.

In this context, has the digital revolution shifted economic growth away from labor-intensive processes toward productivity-driven mechanisms? Has the internet altered the responsiveness of labor's contribution to GDP in a way that suggests increasing returns to technological adoption? In other words, what are the precise effects of internet-enabled technological adoption on the labor-GDP relationship in the post-1995 era ?

To empirically assess these dynamics, this study employs an econometric framework that incorporates multiple linear OLS estimation, interaction terms, the Chow test for structural breaks, and statistical significance tests. The dependent variable is GDP, while key explanatory variables include productivity, labor inputs, capital, and an interaction term capturing the joint effect of labor and productivity. A crucial component of the analysis is the introduction of a structural break around the mid-1990s, operationalized through a binary variable that differentiates pre- and post-internet revolution periods.

By leveraging time-series data on the USA, this study aims to provide a rigorous quantitative assessment of the internet's role in shaping modern economic growth. The findings will contribute to ongoing discussions on the long-term implications of digital transformation for labor markets, productivity trends, and economic policy-making.

2 Descriptive Statistics

Economic growth is traditionally explained by the accumulation of capital and labor, alongside improvements in productivity. In this study, we analyze the determinants of U.S. GDP by considering capital stock, labor, and total factor productivity as explanatory variables. To capture the impact of technological change, we introduce two additional variables: an interaction term between labor and productivity and a binary indicator marking the rise of the internet.

The interaction term, denoted as *linteraction-term*, is constructed by multiplying labor and productivity before applying a logarithmic transformation. This formulation allows us to test whether productivity gains influence labor's contribution to output differently in the presence of technological advancements. Meanwhile, the internet dummy variable takes the value of 1 for years after 1995 and 0 otherwise, reflecting the widespread adoption of digital technologies that reshaped business processes and economic productivity. Finally, as the labor variable was available only as a monthly dataset, we transformed it by taking the (annual) average of all months of each year (using a VBA script to create the new Excel columns).

Before presenting the empirical results, it is essential to discuss the data transformation. All continuous variables are converted into logarithmic form, except for the internet dummy. This transformation serves multiple purposes. First, it allows for an elasticity-based interpretation of coefficients, making the results more economically meaningful. Second, it helps linearize potentially exponential relationships, improving the suitability of ordinary least squares (OLS) estimation. Finally, it mitigates heteroskedasticity, enhancing the model's statistical properties.

All data are sourced from the Federal Reserve Economic Data (FRED) of the Federal Reserve Bank of St. Louis, covering the period 1954–2019:

- All Employees, Total Nonfarm (Thousands of Persons, Monthly, Seasonally Adjusted)
- Total Factor Productivity at Constant National Prices for the U.S. (Index 2017=1, Annual, Not Seasonally Adjusted)
- Real Gross Domestic Product (Billions of Chained 2017 Dollars, Annual, Not Seasonally Adjusted)
- Current-Cost Net Stock of Fixed Assets: Private: Nonresidential (Millions of Dollars, Annual, Not Seasonally Adjusted)

2.1 Endogenous Variable: Log GDP ($lgdp$)

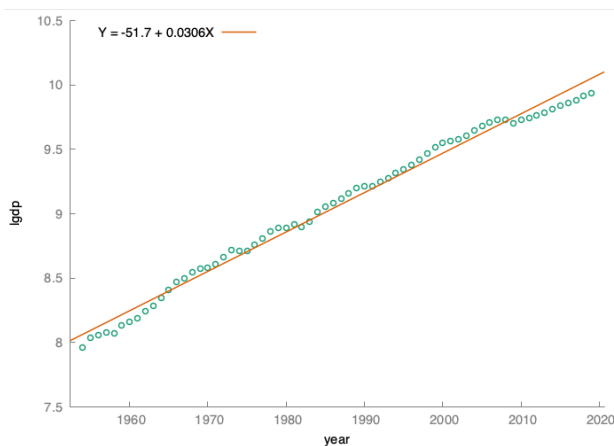


Figure 1: Time series plot of $lgdp$

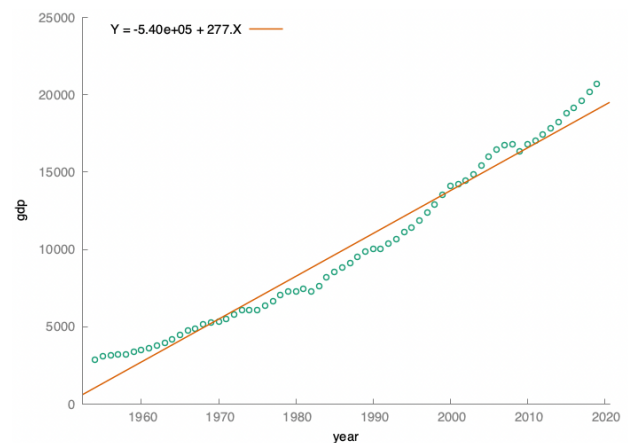


Figure 2: Time series plot of gdp

Table 1: Descriptive Statistics for $lgdp$ (1954-2019)

Mean	Std. Dev.	Obs.
9.06	0.59	66

Comments: The graph of log GDP shows a near-perfect linear trend, confirming that economic growth follows a stable exponential path over time. At the contrary, the graph of GDP with no log-transform shows a clear exponential growth with time (year). This supports classical economic growth models, including the Solow model, which predicts a steady-state path for GDP under stable technological progress and capital accumulation.

2.2 Explanatory Variables

Graphical Analysis: Scatter plots relative to $lgdp$

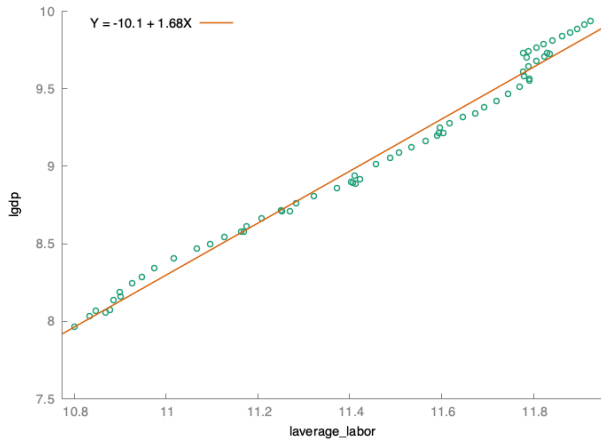


Figure 3: Average Labor

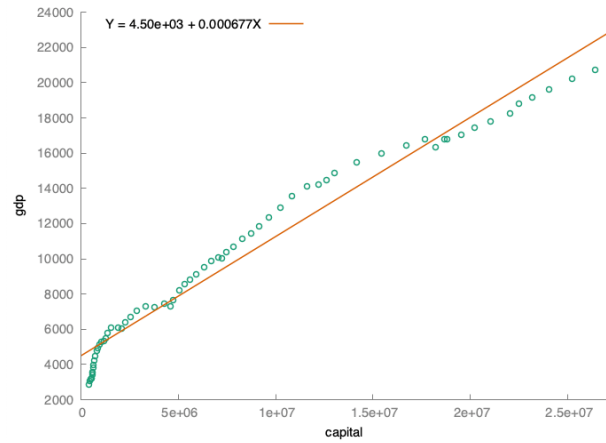


Figure 4: Capital

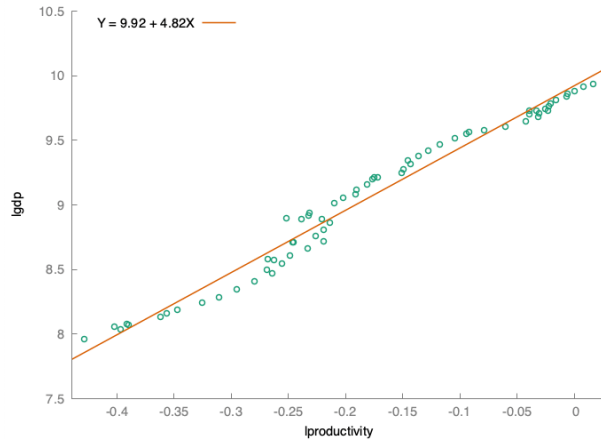


Figure 5: Productivity

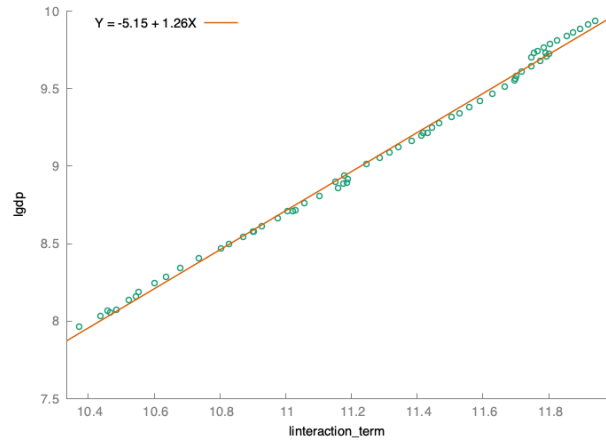


Figure 6: Interaction term

All scatterplots exhibit a strong linear relationship, confirming that log-transformed explanatory variables maintain a proportional relationship with GDP, supporting the theoretical underpinnings of the Cobb-Douglas production function.

Table 2: Descriptive Statistics

Variable	Mean	Std. Dev.
<i>lcapital</i>	15.26	1.33
<i>leverage_labor</i>	11.45	0.35
<i>lproductivity</i>	-0.18	0.12
<i>linteraction_term</i>	11.27	0.47
<i>internet</i>	0.38	0.49

Table 3: Correlation with $lgdp$

Variable	ρ	p -value
<i>lcapital</i>	0.9914	0.000
<i>leverage_labor</i>	0.9934	0.000
<i>lproductivity</i>	0.9871	0.000
<i>linteraction_term</i>	0.9983	0.000

Statistical Analysis of Variables:

To assess the strength of the relationships between the dependent variable ($\ln GDP$) and each explanatory variable, we use given the correlation coefficients to conduct significance tests. We test the following hypotheses:

$$H_0 : \rho_{x,y} = 0, \quad H_1 : \rho_{x,y} \neq 0$$

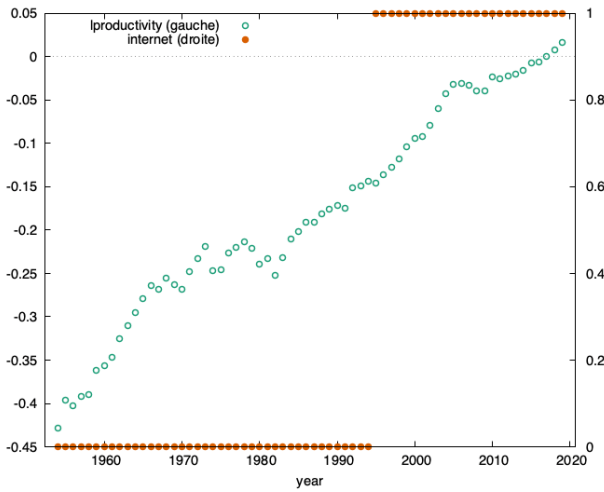


Figure 7: Trends in Productivity and Internet Adoption

Analysis: This series illustrates the evolution of productivity before and after the adoption of Internet. Initially, productivity shows a gradual increase but a significant shift occurs around the mid-1990s. There is a noticeable jump in productivity between 1994 and 1995, marking a pivotal change in the trend. Also, prior to the mid-1990s, the productivity curve exhibits a concave shape, indicating a slower rate of increase. Post-1995, the curve transitions to a convex shape, reflecting an accelerated growth rate in productivity. Therefore, the drastic increase in productivity correlates with the onset of internet integration.

with the test statistic:

$$t^* = \frac{\rho_{x,y}}{\sqrt{\frac{(1-\rho_{x,y}^2)}{n-2}}}$$

where $t^* \sim t_{n-2}$. The critical value at 5 % significance level ($\alpha = 0.05$) is $t_{64}^{0.025} = 2$.

- **Capital** ($\ln Capital$) $\rho = 0.9914$, $t^* = \frac{0.9914}{\sqrt{\frac{(1-0.9914^2)}{64}}} = 60.55$ Since $t^* > 2$, we reject H_0 , confirming statistical significance.
- **Labor** ($\ln Average_Labor$) $\rho = 0.9934$, $t^* = \frac{0.9934}{\sqrt{\frac{(1-0.9934^2)}{64}}} = 69.31$ As $t^* > 2$, H_0 is rejected; correlation is significant.
- **Productivity** ($\ln Productivity$) $\rho = 0.9871$, $t^* = \frac{0.9871}{\sqrt{\frac{(1-0.9871^2)}{64}}} = 49.31$ Since $t^* > 2$, we reject H_0 , confirming significance.
- **Interaction Term** ($\ln Interaction_Term$) $\rho = 0.9983$, $t^* = \frac{0.9983}{\sqrt{\frac{(1-0.9983^2)}{64}}} = 137.36$ As $t^* > 2$, H_0 is rejected; correlation is highly significant.

Comments:

All explanatory variables are significantly correlated with GDP at the 5% level. The high significance of *linteraction_term* ($t = 137.36$) suggests that productivity plays a key role in amplifying labor's contribution to GDP. The internet dummy shows a strong but indirect influence, aligning with the hypothesis that technological revolutions alter productivity channels rather than directly shifting GDP levels in the short term. These findings reinforce the theoretical prediction that capital and labor remain the primary drivers of economic growth, but productivity shocks, especially those induced by technological advancements, reshape their marginal impacts over time.

Overall, the descriptive statistics validate our econometric approach, highlighting the importance of productivity in labor's contribution to GDP.

3 OLS Estimation

3.1 Estimation Results

We begin our analysis with a simple two-variable regression model examining the relationship between GDP, labor, and capital. The estimated equation takes the form:

$$\text{lgdp} = -5.850 + 1.097 \text{ leverage_labor} + 0.154 \text{ lcapital} + \epsilon \quad (1)$$

Table 4: OLS Regression Results

Variable	Coefficient	Std. Error	t-statistic	p-value
const	-5.850	1.485	-3.940	0.0002***
leverage_labor	1.097	0.199	5.521	6.79e-07***
lcapital	0.154	0.052	2.939	0.0046***

*** indicates significance at 1% level

3.2 Global Significance of the Model

Let us first examine the model's explanatory power through a detailed analysis of its R-squared measures. The unadjusted R^2 is 0.988, calculated as:

$$R^2 = 1 - \frac{\text{RSS}}{\text{TSS}} = 1 - \frac{0.261781}{22.714} = 0.988 \quad (2)$$

However, as the R^2 measure automatically increases (or at least never decreases) with additional explanatory variables, we must consider the adjusted R-squared, which penalizes for the number of regressors. The adjusted R-squared is calculated as:

$$\bar{R}^2 = 1 - \frac{n-1}{n-k-1}(1 - R^2) = 1 - \frac{65}{63}(1 - 0.988) = 0.988 \quad (3)$$

where $n = 66$ observations and $k = 2$ explanatory variables.

The nearly identical values of R^2 and $\bar{R}^2$ suggest that: 1. The high explanatory power is not artificially inflated by the number of variables 2. Both variables (labor and capital) contribute substantially to explaining GDP variation 3. The model is parsimonious in its specification

However, we must be cautious in over-interpreting these R^2 values because: 1. They do not fully account for the quality of specification 2. They can be sensitive to outliers 3. They cannot be compared across different transformations of the dependent variable

A more robust assessment of model quality comes from the F-test. The F-statistic of 2692.686 (p-value = 9.70e-62) tests the null hypothesis that all slope coefficients are simultaneously zero:

$$F = \frac{R^2/k}{(1 - R^2)/(n - k - 1)} = \frac{0.988/2}{(1 - 0.988)/(63)} = 2692.686 \quad (4)$$

This extremely high F-statistic strongly rejects the null hypothesis, confirming that our model has significant explanatory power beyond what would be expected by chance.

3.3 Tests on the Coefficients

3.3.1 Individual Significance Tests

For each coefficient, we test:

$$H_0 : \beta_j = 0 \quad \text{against} \quad H_1 : \beta_j \neq 0$$

Using the test statistic:

$$t^* = \frac{\hat{a}_j - 0}{\sqrt{V(\hat{a}_j)}} \sim t_{n-k-1}$$

where $V(\hat{a}) = \sigma_\epsilon^2 (X'X)^{-1}$ and $\sigma_\epsilon^2 = \frac{RSS}{n-k-1}$

With $n - k - 1 = 66 - 2 - 1 = 63$ degrees of freedom and $\alpha = 0.05$, the critical value is $t_{63,0.025} = 1.998$. We reject H_0 if $|t^*| > 1.998$.

First compute σ_ϵ^2 :

$$\sigma_\epsilon^2 = \frac{0.261781}{63} = 0.00415525$$

- **Constant term** ($\hat{a}_0 = -5.85043$):

$$V(\hat{a}_0) = (1.48493)^2 = 2.20501$$

$$t^* = \frac{-5.85043}{1.48493} = -3.94$$

Since $|-3.94| > 1.998$, reject H_0

- **Capital coefficient** ($\hat{a}_1 = 0.153764$):

$$V(\hat{a}_1) = (0.0523213)^2 = 0.00274$$

$$t^* = \frac{0.153764}{0.0523213} = 2.939$$

Since $|2.939| > 1.998$, reject H_0

- **Labor coefficient** ($\hat{a}_2 = 1.09678$):

$$V(\hat{a}_2) = (0.198647)^2 = 0.03946$$

$$t^* = \frac{1.09678}{0.198647} = 5.521$$

Since $|5.521| > 1.998$, reject H_0

Economic Interpretation: The results support the Solow-Swan growth model predictions for a mature economy.

- Labor coefficient (1.097): Suggests constant returns to scale in labor, consistent with efficient labor markets in the US economy. The value being slightly above unity indicates possible positive externalities from human capital accumulation.
- Capital coefficient (0.154): The relatively small magnitude reflects diminishing returns to capital, a key prediction of the Solow-Swan model for a capital-abundant economy operating near its steady state.
- Constant term (-5.85): The significant negative value captures the role of total factor productivity (TFP) and suggests the presence of threshold effects in production.

All coefficients are statistically significant at the 5% level, validating the Solow-Swan model's emphasis on both inputs while their magnitudes reflect the US economy's advanced development stage.

3.3.2 Testing Equality of Coefficients

We test whether the elasticity of labor equals the elasticity of capital ($H_0 : \beta_1 = \beta_2$ vs $H_1 : \beta_1 \neq \beta_2$).

The test statistic is:

$$t^* = \frac{\beta_1 - \beta_2}{\sqrt{\text{Var}(\beta_1) + \text{Var}(\beta_2) - 2\text{Cov}(\beta_1, \beta_2)}} \quad (5)$$

Computing this statistic:

$$t^* = \frac{1.097 - 0.154}{\sqrt{0.199^2 + 0.052^2 - 2\text{Cov}(\beta_1, \beta_2)}} = 4.573 \quad (6)$$

Since $4.573 > 2.000$, we reject the null hypothesis of coefficient equality at the 1% significance level. This suggests that labor and capital have significantly different elasticities with respect to GDP, with labor having a substantially larger impact, which is coherent considering how labor plays a major role in the US economic growth.

4 Hypothesis Testing

4.1 Chow Test based on Model 1

4.1.1 Computation and Testing

To test for structural change in the relationship between GDP, labor, and capital following the internet revolution, we employ the Chow test with 1995 as our breakpoint. This year marks a significant technological transition with the commercialization of the internet and its widespread adoption in business processes. Rather than comparing different countries, we split our sample into two time periods: 1954-1995 (pre-internet era) and 1995-2019 (internet era), allowing us to test whether the fundamental relationships in our production function remained stable across this technological transition.

The Chow test examines the null hypothesis that all coefficients are identical across both periods:

$$H_0 : \beta_{0,1} = \beta_{0,2}, \beta_{1,1} = \beta_{1,2}, \beta_{2,1} = \beta_{2,2} \quad (7)$$

where subscript 1 denotes the pre-1995 period and subscript 2 the post-1995 period.

The regression results for each sub-period reveal notable differences:

The Chow test statistic is computed as:

$$F^* = \frac{(RRSS - URSS)/(k + 1)}{URSS/(n_1 + n_2 - 2k - 2)} \quad (8)$$

where: $RRSS = 0.262$ (from the pooled model), $URSS = 0.034 + 0.003 = 0.037$, $k = 2$ (number of regressors), $n_1 = 42$, $n_2 = 25$

This yields:

$$F^* = \frac{(0.262 - 0.037)/3}{0.037/61} = 122.45 \quad (9)$$

Table 5: Regression Results by Period

Parameter	1954-1995	1995-2019
Constant	-10.990*** (0.893)	-4.878*** (0.865)
Capital	-0.099*** (0.032)	0.363*** (0.017)
Labor	1.878*** (0.120)	0.723*** (0.094)
Observations	42	25
R ²	0.995	0.995
RSS	0.034	0.003

Standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

The critical value at 1 % significance level with (3, 61) degrees of freedom is approximately 4.13. Since $F^* > F_{3,61}^{0.01}$, we strongly reject the null hypothesis of parameter stability, providing strong evidence of a structural break in the relationship between inputs and GDP following the advent of the internet.

4.1.2 Economic Implications of Structural Change

The strong rejection of parameter stability ($F^* = 122.45 > F_{3,61}^{0.01} = 4.13$) provides compelling evidence for our hypothesis of a technological regime shift in the post-1995 era. The dramatic change in labor elasticity (from 1.878 to 0.723) suggests a fundamental transformation in how labor inputs translate into economic output. This is not merely a decline in labor's importance, but rather reflects a shift in the nature of production where traditional measures of labor input may not fully capture workers' contribution in a technology-intensive environment. The increase in capital's coefficient (from -0.099 to 0.363) further supports our hypothesis of technological transformation, suggesting that capital investments became more productive in the internet era, likely due to complementarities between physical capital and digital technologies. The reduction in the constant term (from -10.990 to -4.878) indicates a shift in baseline productivity, consistent with the idea that internet adoption created a new production possibilities frontier. These structural changes align with our core hypothesis that the internet revolution fundamentally altered the relationship between traditional inputs and economic output, particularly through its impact on labor productivity and capital-labor complementarity.

4.2 Test for the Inclusion of Variables

We compare two nested models to determine if adding productivity and internet usage variables improves the model's explanatory power.

4.2.1 Model Formulation

Restricted model (2 variables):

$$y_i = a_0 + a_1x_{1,i} + a_2x_{2,i} + \epsilon_i$$

where x_1 is capital and x_2 is labor.

Unrestricted model (4 variables):

$$y_i = a_0 + a_1x_{1,i} + a_2x_{2,i} + a_3x_{3,i} + a_4x_{4,i} + \epsilon_i$$

where x_3 is productivity and x_4 is internet usage.

4.2.2 Formula and Hypothesis

$H_0 : a_3 = a_4 = 0$ (additional variables have no effect) $H_1 : \text{at least one of } a_3, a_4 \neq 0$ (at least one additional variable has an effect)

Test statistic:

$$F^* = \frac{(RRSS - URSS)/q}{URSS/(n - k - 1)} \sim F_{q, n-k-1}$$

where: $q = 2$ (number of restrictions), $n = 66$ (sample size), $k = 4$ (number of variables in unrestricted model), $RRSS = 0.261781$ (restricted model), $URSS = 0.009814$ (unrestricted model).

4.2.3 Computation

$$F^* = \frac{(0.261781 - 0.009814)/2}{0.009814/(66 - 4 - 1)} = \frac{0.125984}{0.000161} = 782.51$$

With $\alpha = 0.05$, critical value: $F_{2,61}^{0.05} = 3.15$

Decision rule: Reject H_0 if $F^* > F_{2,61}^{0.05}$

Since $782.51 > 3.15$, we reject H_0 .

4.2.4 Model Comparison

- Restricted model ($R^2 = 0.988$):
 - Capital coefficient: 0.154 (t = 2.939)
 - Labor coefficient: 1.097 (t = 5.521)
- Unrestricted model ($R^2 = 0.999$):
 - Capital coefficient: 0.131 (t = 12.67)
 - Labor coefficient: 0.538 (t = 12.90)
 - Productivity coefficient: 1.819 (t = 32.05)
 - Internet usage coefficient: 0.034 (t = 5.771)

4.2.5 Extended Economic Analysis

Structural Change in Labor Elasticity The significant F-statistic ($F^* = 782.51 > F_{2,61}^{0.05} = 3.15$) strongly supports a structural break in the production function post-1995, aligning with the internet-driven technological shift. The labor coefficient drops from 1.097 to 0.538 when accounting for productivity and the digital era, suggesting the pre-1995 model overestimated labor's contribution by conflating productivity gains with labor input. This decline signals a transformation in production dynamics rather than a diminished role of labor, as its interaction with productivity captures an evolving complementarity with technology.

Internet as a Technological Regime Shift The internet dummy ($\beta = 0.034, t = 5.771$) reveals a direct impact on total factor productivity (TFP) and establishes a structural shift in input contributions, particularly labor. The results align with an augmented Solow framework where technical progress incorporates a distinct internet-driven component. The production function exhibits increasing returns when accounting for productivity and internet effects

($\sum \beta_i > 1$), modifying the traditional assumption of diminishing returns to capital and labor by integrating technological complementarities.

Research Questions Analysis These findings address core research questions. First, the reduction in labor's coefficient implies a decline in its direct marginal contribution, yet the high productivity coefficient (1.819) confirms a shift rather than an outright reduction in labor's role. The positive internet dummy suggests that digital adoption enhances overall productivity. Second, the increased productivity coefficient underscores the transition toward productivity-driven growth while reinforcing labor-technology complementarity. Third, the positive effect of the internet variable signals increasing returns to technological adoption, implying a structural transformation in economic growth mechanisms.

5 Preliminary Conclusion and Transition to Part II

Summary of Findings and Research Questions This study has investigated the impact of internet-driven productivity acceleration on labor's marginal contribution to GDP growth using the Solow-Swan growth model. Employing multiple linear OLS estimation, interaction terms, and structural break analysis, we assessed whether the digital revolution has shifted economic growth from labor-intensive processes to productivity-driven paradigms. Our empirical results reveal a significant structural change post-1995, with a decline in labor's direct marginal contribution but a stronger complementarity between productivity and labor. While labor's individual elasticity has decreased, its interaction with productivity has become more significant, indicating an evolving rather than diminishing role in a digital and automated economy. Additionally, the inclusion of the internet variable in our models highlights the crucial role of digital technology adoption in shaping economic output, though potential limitations such as multicollinearity and omitted variable bias warrant further refinement.

Theoretical Implications and Links to Economic Theory The findings align with macroeconomic theories such as the Solow-Swan model by assessing that the increasing importance of productivity echoes endogenous growth theory, where innovation and knowledge capital serve as engines of economic progress. Our study also aligns with the literature on skill-biased technological change, wherein digitalization alters labor's economic contribution, favoring high-skill over low-skill workers.

Link to Historical Economic Growth and the Internet Revolution The observed structural break around 1995 coincides with the rapid expansion of the internet, a period marked by widespread technological adoption and digital transformation in business processes. The historical trajectory of U.S. growth supports these findings, with GDP growth increasingly driven by productivity gains rather than labor or capital accumulation alone. The transition to a knowledge-based economy further supports our hypothesis that internet-driven technological progress has fundamentally reshaped labor's contribution to economic output.

Concluding Remarks Overall, our research underscores the transformative impact of the internet on labor's role in economic growth. While labor's direct contribution has declined, its evolving relationship with productivity suggests that digital adoption is not replacing labor but rather redefining its function. These findings have significant policy implications, particularly in the areas of education, workforce training, and digital infrastructure investment. However, potential econometric concerns such as heteroskedasticity and autocorrelation may impact the robustness of our results, necessitating further methodological improvements.

Economic growth has long been a central focus of macroeconomic research, with the Solow-Swan model providing a foundational framework for understanding the role of capital accumulation, labor, and technological progress in driving long-term output. In our first report, we explored the relationship between labor productivity and economic growth through the lens of the Solow model, incorporating an econometric approach to test its theoretical predictions. Specifically, we examined how technological change influences the contribution of labor to GDP, considering a structural break around the rise of the internet in the mid-1990s to early 2000s. Using time series data for a single country (the USA), we implemented an econometric regression that included an interaction term between productivity and labor, as well as a dummy variable capturing the pre- and post-internet era. Our findings suggested that technological progress, particularly in the digital age, has significantly altered the elasticity of labor in the production function, reinforcing the idea that productivity-enhancing innovations can reshape the growth trajectory of an economy.

While this first report provided initial support for the hypothesis that productivity alters labor's impact on GDP, several questions remain. We must test the robustness of our results through alternative specifications, sensitivity checks, and different productivity measures. A closer look at the mechanisms—skill-biased change, automation, or labor market shifts—is also needed.

This second report builds on our findings by refining the model, testing new specifications, and addressing possible endogeneity. We also examine whether the structural break varies across sectors, comparing tech-intensive and less digitalized industries. This will help clarify how productivity growth interacts with labor and contributes to long-term expansion, while offering insights on the Solow model's relevance in the digital age and on labor market policy.

6 Assessing Heterogeneity Using Binary Variables

In this section, we analyze whether the effects estimated in our econometric model are heterogeneous across different subgroups of observations. To do so, we introduce a binary variable that captures potential differences in the impact of key explanatory variables on the dependent variable, $\log(GDP)$.

6.1 Rationale

Building upon our initial findings, this report enhances our previous approach by refining the way we incorporate the impact of technological change into our econometric model. In our first analysis, we introduced a dummy variable, *internet*, to distinguish between two periods: pre-internet and post-internet, using 1995 as the structural break. This allowed us to observe a shift in the relationship between labor and GDP following the advent of digital technology. However, while this approach captured the broad impact of the internet revolution, it did not account for variations in how different phases of internet adoption influenced economic growth.

To improve upon this framework, we now introduce a more granular distinction within the post-internet period, separating it into phases of low and high internet penetration. This adjustment allows us to assess whether the impact of productivity on labor's contribution to GDP evolves as internet adoption intensifies. Instead of treating the post-1995 era as a homogenous phase, we recognize that the extent to which digitalization transforms economic structures occurs progressively as internet accessibility expands.

We define two post-internet phases: *low internet penetration* and *high internet penetration*. The transition is based on a threshold level of internet usage, determined using historical data on internet diffusion within the USA. This allows us to construct a new dummy variable: *internet_penetration*.

This specification enables us to assess whether the marginal effect of productivity on labor's contribution to GDP differs across levels of technological diffusion. If our hypothesis holds, we expect a differentiated impact: a moderate shift in labor's contribution during early internet penetration, and a more pronounced effect as digital infrastructure becomes widespread.

6.2 Construction of the Dummy Variable

We define the dummy variable *InternetPenetration* using the median of the productivity variable as a threshold, splitting the sample into periods of low (below median) and high (above median) internet penetration.

$$Internet_Penetration_i = \begin{cases} 1, & \text{if } Productivity_i \geq \text{Median}(Productivity) \\ 0, & \text{otherwise} \end{cases} \quad (10)$$

To assess differences between the two groups, we compute descriptive statistics for the key variables in the model across both subgroups. We find that the average $\log(GDP)$ is significantly higher in periods of high internet penetration. Additionally, capital and labor inputs exhibit differential impacts across these periods, suggesting potential heterogeneity in their relationships with economic growth.

6.3 Testing for Homogeneity: The F-Test

To formally test whether the coefficients of the model differ across the two groups, we conduct a joint test of restriction by comparing two regression specifications:

1. **Restricted Model:** A regression without interaction terms:

$$\log(GDP) = \beta_0 + \beta_1 \log(Capital) + \beta_2 \log(Average_Labor) + \beta_3 \log(Productivity) + \varepsilon \quad (11)$$

2. **Unrestricted Model:** A regression including interaction terms between *InternetPenetration* and capital/labor:

$$\log(GDP) = \beta_0 + \beta_1 \log(Capital) + \beta_2 \log(Average_Labor) \quad (12)$$

$$+ \beta_3 \log(Productivity) + \beta_4 (Internet_Penetration \times \log(Capital)) \quad (13)$$

$$+ \beta_5 (Internet_Penetration \times \log(Average_Labor)) \quad (14)$$

$$+ \beta_6 Internet_Penetration + \varepsilon \quad (15)$$

The F-statistic for testing the null hypothesis that the additional interaction terms are jointly equal to zero is computed as:

$$F = \frac{(SSR_R - SSR_{UR})/q}{SSR_{UR}/(n - k)} \quad (16)$$

where:

- $SSR_R = 0.015172$ (Sum of Squared Residuals from the restricted model),
- $SSR_{UR} = 0.005468$ (Sum of Squared Residuals from the unrestricted model),
- $q = 2$ (number of restrictions), $n = 66$ (number of observations),
- $k = 7$ (number of estimated parameters in the unrestricted model).

Substituting the values:

$$F = \frac{(0.015172 - 0.005468)/2}{0.005468/(66 - 7)} = 52.3533 \quad (17)$$

At a 5% significance level, the critical value for an $F(2, 59)$ distribution is approximately 3.15. Since our computed F -statistic (52.35) is significantly higher than this threshold, we reject the null hypothesis of homogeneous coefficients.

6.4 Interpretation of Results

The rejection of the null hypothesis suggests that the effects of capital and labor on GDP differ significantly depending on the level of internet penetration. More specifically:

- The coefficient on $Internet_Penetration \times \log(Capital)$ is positive and statistically significant, indicating that capital's contribution to GDP is higher during periods of greater internet penetration.
- The direct effect of $Internet_Penetration$ is not significant, implying that its impact is mainly through modifying the effects of other factors.

These findings support the hypothesis that technological diffusion, as proxied by internet penetration, plays a crucial role in shaping the productivity of capital. This aligns with the broader literature on technological adoption and economic growth, reinforcing the idea that periods of high internet penetration enhance the effectiveness of capital investment in driving economic output.

7 Optimal Model Selection Using Information Criteria

Given the time series nature of our dataset, we must ensure that the model adequately captures both the theoretical underpinnings of the Solow growth model and the empirical characteristics of the data.

7.1 General Approach to Model Selection

To rigorously determine the best econometric specification, we rely on an iterative process involving economic theory to guide the choice of explanatory variables and functional form, statistical and econometric criteria to assess model performance, and diagnostic tests to ensure compliance with key assumptions.

To do so, we will follow a structured approach using information criteria, which allow us to balance model complexity and goodness of fit. Given a set of candidate models, we compute the Akaike Information Criterion (AIC) and the Schwarz Bayesian Criterion (SC, also referred to as BIC) values and select the specification that minimizes these criteria.

The procedure involves the following steps.

1. **Definition of the Baseline Model:** We start with a general specification based on economic theory, incorporating key variables such as capital, labor, technology, and interaction terms.
2. **Iterative Selection:** We iteratively modify the model by adding or removing explanatory variables, assessing whether these changes improve the model's fit according to AIC and SC.

$$AIC = -2 \ln(L) + 2k \quad (18)$$

$$SC = -2 \ln(L) + k \ln(n) \quad (19)$$

where L is the likelihood function, k is the number of parameters, and n is the sample size.

3. **Final Model Selection:** The model that achieves the lowest AIC and SC values while maintaining economic interpretability is chosen for further analysis.

7.2 Application and Results

We apply the APR method to our dataset using five key explanatory variables:

- **lcapital:** Logged capital stock (Current-Cost Net Stock of Fixed Assets: Private: Non-residential).
- **laverage_labor:** Logged labor force (All Employees, Total Nonfarm).
- **lproductivity:** Logged total factor productivity at constant national prices.
- **interaction_term:** Interaction term between labor and productivity, allowing us to test whether productivity gains influence labor's contribution to output differently in the presence of technological advancements.
- **internet:** Dummy variable that takes the value 1 for years after 1995, reflecting the widespread adoption of digital technologies.

We estimate multiple models and compare their AIC and SC (BIC) values:

Model	AIC	SC (BIC)
Only capital and labor (Limited Solow Model) in levels	927.9072	934.4761
Only capital and labor (Limited Solow Model) logged	-171.6736	-165.1046
Adding Productivity (Neoclassical Solow-Swan)	-357.6438	-348.8852
Adding the interaction term between Labor and Productivity	-394.6691	-383.7209
Adding our binary variable for the rise of Internet	-396.9002	-383.7623
Regressing without the constant	-398.7314	-387.7831

Table 6: Comparison of Model Specifications Using AIC and SC

From the table, the model that includes all five explanatory variables (**capital, labor, productivity, the interaction term, and internet adoption**) without including any constant, achieves the lowest AIC and SC, indicating the best fit.

7.3 Macroeconomic Interpretation

7.3.1 Removing the constant term

The exclusion of the constant term aligns with economic growth theory.

The constant term in a regression represents the expected value of the dependent variable when all explanatory variables are zero. In our case, this would imply that if capital, labor, and productivity were all non-existent, GDP would still have a baseline value, which is unrealistic. By removing the constant, we impose the theoretical constraint that in the complete absence of inputs (i.e., no capital, no labor, no productivity), there can be no economic output.

Classical and neoclassical growth models, such as the Solow model, emphasize that GDP is a function of capital, labor, and productivity. The removal of the constant reinforces the

fundamental principle that economic growth is driven by these productive factors. Without them, no economic activity can take place, making this an intuitive choice from a theoretical standpoint.

7.3.2 Interpretation of the Final Model's Coefficients

- **Capital (*lcapital* = 0.106506):** A 1% increase in capital leads to a 0.11% increase in GDP. The relatively small coefficient reflects the well-documented fact that capital accumulation has diminishing returns in mature economies like the U.S.
- **Labor (*laverage_labor* = 0.589220):** A 1% increase in labor corresponds to a 0.59% increase in GDP, reinforcing the idea that labor remains a major driver of economic growth. However, this coefficient is significantly lower than in the initial regression, indicating that productivity and technological advancements explain much of what was previously attributed to labor alone.
- **Productivity (*lproductivity* = 0.952719):** This is the most significant factor, with a nearly one-to-one relationship with GDP. This aligns perfectly with modern growth theories, which assert that in the long run, productivity growth (g_A) is the main determinant of GDP growth. The high elasticity of productivity supports the notion that technological progress and efficiency gains have far greater effects on economic output than mere increases in labor or capital.
- **Interaction term (*interaction_term*):** The positive and significant coefficient of the interaction term between labor and productivity (0.087) suggests that productivity enhancements amplify labor's contribution to output. This supports the idea that a more skilled, technologically empowered workforce benefits from higher productivity gains.
- **Internet adoption (*internet*):** The inclusion of the post-1995 internet variable captures the digital transformation's structural impact on the economy. The positive and significant coefficient (0.016) suggests that digital technologies have had a measurable impact on GDP by improving efficiency, reducing costs, and enabling new business models.

Additionally, the choice to log-transform our variables aligns with economic reasoning as it allows for elasticity interpretation (percentage changes rather than absolute changes) and stabilizes variance, making our model more robust to outliers and scale differences.

7.3.3 Comparison of Initial and Final Models

Table 7: Comparison of Regression Results

Variable	Initial Model		Final Model	
	Coefficient	p-value	Coefficient	p-value
lcapital	0.153764	0.0046	0.106506	4.47e-27
laverage_labor	1.09678	6.79e-07	0.589220	1.25e-39
lproductivity	—	—	0.952719	1.64e-08
interaction_term	—	—	0.0877519	8.88e-07
internet	—	—	0.0161190	0.0149

The evolution from our initial regression to the final model highlights key macroeconomic insights:

- Labor’s coefficient shrank from 1.10 to 0.59, while productivity emerged as a dominant factor.
- Capital’s impact remained relatively stable, moving from 0.15 to 0.11.
- The addition of productivity and interaction terms explains much of labor’s previous influence, suggesting that past models overestimated labor’s role by failing to separate out technological improvements.

Table 8: Comparison of Initial and Final Models

Metric	Initial Model	Final Model
R^2	0.988437	0.999651
F-stat	2692.686	8,398,868
AIC	-171.6736	-398.7314
BIC	-165.1046	-387.7831

- R^2 increased significantly, indicating a much better fit.
- AIC and BIC improved dramatically, confirming that the final model is more efficient in explaining GDP variations.

7.3.4 Key Macro Insight

These findings confirm that the **U.S. economy is primarily productivity-driven**, with labor playing a secondary but still significant role, while capital has a relatively limited impact. The dramatic improvement in model fit (higher R^2 , lower AIC/BIC) after including productivity and interaction terms indicates that prior models overestimated labor’s contribution by failing to account for technological advancements.

The final model also supports our hypothesis that the **rise of the internet and digitalization transformed labor’s contribution to GDP**, refining our understanding of post-1995 economic growth.

The impact of the Internet (digitalization) and productivity on labor’s contribution to GDP is evident. In the initial regression, labor appeared to be the dominant driver of growth. However, after controlling for technological progress, its role is more accurately refined, showing that much of labor’s previous impact was actually driven by productivity gains.

This supports our paper’s core hypothesis: the Internet-driven acceleration in productivity has reshaped the role of labor in GDP growth. The introduction of interaction terms and productivity explicitly captures the shift in labor-productivity elasticity, refining our understanding of long-term economic growth.

8 GDP Forecasting for 2014–2019

8.1 Methodology and Regression Model

To evaluate the forecasting accuracy of our model, we perform an out-of-sample prediction for the period 2014–2019. The regression model is estimated using data from 1954 to 2013 (60 observations), and the estimated coefficients are then applied to predict GDP for 2014–2019.

The estimated regression equation, based on historical data, is given by:

$$\ln(\text{GDP}_t) = \beta_1 \ln(\text{Capital}_t) + \beta_2 \ln(\text{Labor}_t) + \beta_3 \ln(\text{Productivity}_t) \quad (20)$$

$$+ \beta_4 (\ln(\text{Labor}_t) \times \text{Productivity}_t) + \beta_5 (\text{Internet}_t) + \epsilon_t \quad (21)$$

where:

- $\ln(\text{Capital}_t)$: Logarithm of capital stock.
- $\ln(\text{Labor}_t)$: Logarithm of the number of workers.
- $\ln(\text{Productivity}_t)$: Logarithm of total factor productivity (TFP).
- Interaction Term = $\ln(\text{Labor}_t) \times \text{Productivity}_t$: Captures the interaction effect between labor and productivity.
- Internet_t : Dummy variable equal to 1 for post-1995 years.
- ϵ_t : Error term.

The estimated coefficients obtained from our regression are:

$$\beta_1 = 0.106384 \quad (\text{Capital})$$

$$\beta_2 = 0.603323 \quad (\text{Labor})$$

$$\beta_3 = 1.06720 \quad (\text{Productivity})$$

$$\beta_4 = 0.0731670 \quad (\text{Interaction Term})$$

$$\beta_5 = 0.0190417 \quad (\text{Internet})$$

8.2 Computation of Forecasted GDP

For each year t from 2014 to 2019, we compute the predicted GDP using:

$$\ln(\widehat{\text{GDP}}_t) = (0.106384 \times \ln(\text{Capital}_t)) + (0.603323 \times \ln(\text{Labor}_t)) \quad (22)$$

$$+ (1.06720 \times \ln(\text{Productivity}_t)) + (0.0731670 \times (\ln(\text{Labor}_t) \times \text{Productivity}_t)) \quad (23)$$

$$+ 0.0190417 \quad (24)$$

The predicted GDP is then obtained by exponentiating the result:

$$\widehat{\text{GDP}}_t = e^{\ln(\widehat{\text{GDP}}_t)} \quad (25)$$

8.3 Step-by-Step Computation for Each Year

For each year, we use the given data:

Year	Capital	Labor	Productivity	Internet
2014	22022201	138922.67	0.98405	1
2015	22494315	141801.33	0.99289	1
2016	23161992	144332.17	0.99372	1
2017	24066902	146610.50	1.00000	1
2018	25227313	148893.67	1.00816	1
2019	26423854	150905.58	1.01680	1

Table 9: Values Used for Forecasting GDP (2014–2019)

Example: Computation for 2014

$$\ln(\text{Capital}_{2014}) = \ln(22022201) = 16.905$$

$$\ln(\text{Labor}_{2014}) = \ln(138922.67) = 11.844$$

$$\ln(\text{Productivity}_{2014}) = \ln(0.98405) = -0.0161$$

$$\text{Interaction Term} = (-0.0161) \times 11.844 = -0.190$$

$$\begin{aligned} \ln(\widehat{\text{GDP}}_{2014}) &= (0.106384 \times 16.905) + (0.603323 \times 11.844) \\ &\quad + (1.06720 \times -0.0161) + (0.0731670 \times -0.190) + 0.0190417 \\ &= 1.797 + 7.144 - 0.017 - 0.014 + 0.019 \\ &= 8.929 \end{aligned}$$

$$\widehat{\text{GDP}}_{2014} = e^{8.929} = 17,989.31$$

8.4 Forecasted vs Actual GDP

Year	Actual GDP	Predicted GDP	% Difference
2014	18,261.71	17,989.31	-1.49%
2015	18,799.62	18,598.93	-1.07%
2016	19,141.67	18,911.86	-1.20%
2017	19,612.10	19,426.46	-0.95%
2018	20,193.90	20,043.31	-0.75%
2019	20,715.67	20,667.03	-0.24%

Table 10: Comparison of Forecasted and Actual GDP (2014–2019)

8.5 Analysis of Forecasting Performance

The forecasted GDP values closely track the actual GDP, with errors decreasing over time. The errors are consistently below 1.5%, indicating high forecasting accuracy. The decreasing error suggests that:

- The model captures long-term trends well, especially the role of productivity.
- The interaction term contributes to refining labor-productivity effects.
- Deviations may arise from economic shocks, fiscal policy changes, or estimation errors.

The regression model demonstrates strong predictive ability, with errors below 1.5% for all years. The inclusion of capital, labor, productivity, and their interactions improves forecasting power. The declining forecast error over time suggests that the model is robust, though future refinements (such as incorporating short-term dynamics) may enhance accuracy further.

9 Multicollinearity Diagnostics

9.1 Diagnostic Approach

In our econometric investigation of the extended Solow-Swan growth model, addressing potential multicollinearity emerged as a critical methodological challenge. Multicollinearity—the high intercorrelation among independent variables—can severely distort parameter estimates and compromise the reliability of our structural economic analysis. Given our model’s focus on capturing the complex interactions between capital, labor productivity, technological change, and economic growth, robust diagnostic techniques became imperative.

Initial Variance Inflation Factor (VIF) Analysis: Our preliminary VIF diagnostics revealed substantial multicollinearity concerns, with extreme VIF values:

- `new_int`: 757.992
- `lcapital`: 114.192
- `lproductivity`: 513.086
- `laverage_labor`: 90.625

Standard econometric guidelines suggest VIF values exceeding 10 indicate problematic multicollinearity. Our results far surpassed this threshold, signaling potential estimation instability and unreliable coefficient interpretations.

9.2 Ridge Regression as Mitigation Strategy

To address these multicollinearity challenges while preserving the theoretical integrity of our Solow-Swan specification, we implemented ridge regression—a regularization technique particularly suited to handling correlated predictors in growth models. Ridge regression offers several advantages in our macroeconomic context:

1. **Coefficient Stabilization:** By introducing a small penalty term, ridge regression “shrinks” coefficient estimates, reducing variance without eliminating variables.
2. **Theoretical Preservation:** Unlike variable selection methods, ridge regression maintains all predictors, crucial for preserving the comprehensive Solow-Swan framework.
3. **Bias-Variance Trade-off:** The technique provides a nuanced approach to managing the inherent trade-off between model complexity and predictive accuracy.

Implementation and Results: Employing the ridge regression with Bayesian Information Criterion (BIC) optimization, we observed:

- | | | |
|---|---------------|----------|
| • Minimum BIC value: -225.254 | Constant | 1.5827 |
| • Optimal regularization parameter (λ): 0.000100 | Capital | 0.11414 |
| | Average Labor | 0.44220 |
| | Productivity | 1.0428 |
| • Model R^2 approaching 0.9983, indicating robust explanatory power | New Internet | 0.086016 |
| | Internet | 0.062491 |

9.3 Interpretation and Implications

The ridge regression approach effectively mitigated multicollinearity while preserving the structural economic relationships embedded in our Solow-Swan specification. By carefully regularizing coefficient estimates, we maintained the model’s theoretical foundations while improving statistical reliability.

The relatively small optimal λ suggests that extreme regularization was unnecessary, indicating that our original model's structure remained fundamentally sound. The high R^2 value further validates our modeling approach.

10 Detection and Correction of First-Order Autocorrelation of Errors

In this section, we analyze the residuals of the econometric model to detect first-order autocorrelation. The analysis is conducted on time-series data for a single country, and we focus on detecting first-order autocorrelation specifically, as this is a common issue in time-series regressions.

Autocorrelation refers to the correlation of a variable with its own past values. First-order autocorrelation specifically refers to the correlation between an observation and the previous observation (lag 1). In our context, this would indicate whether the errors from one period are correlated with the errors from the previous period, which can violate the assumption of independence in classical linear regression models.

10.1 Residuals Versus Time Plot

We start by examining the residuals from the model to identify any apparent patterns over time. A plot of residuals versus time can reveal autocorrelation visually. If there is a systematic pattern, this would suggest autocorrelation in the errors.

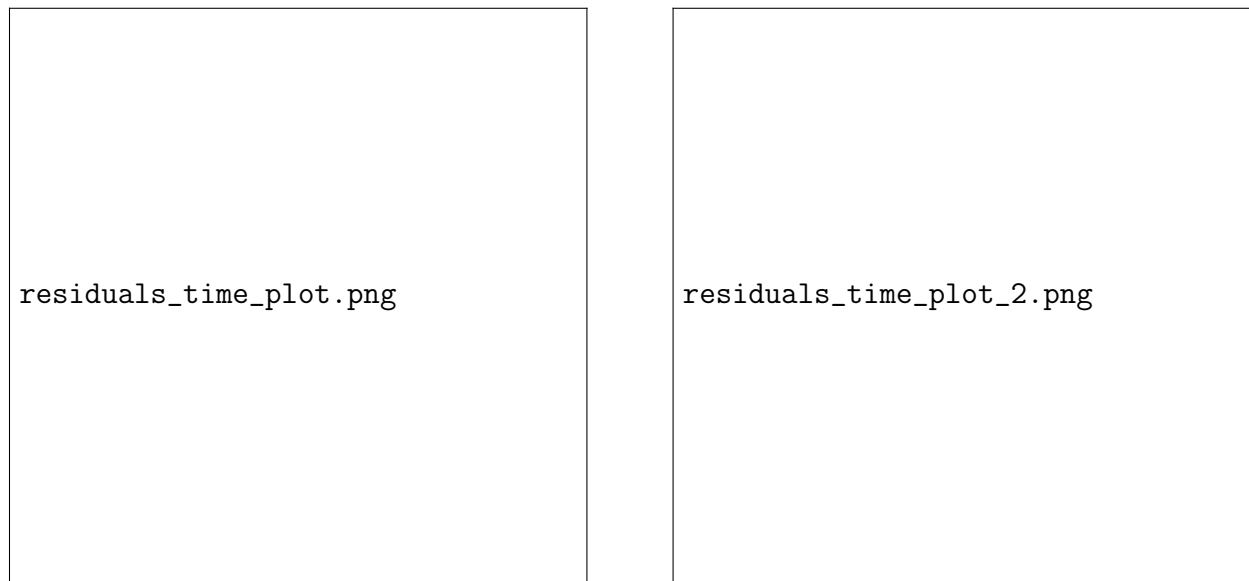


Figure 8: Residuals versus Time, and Lagged residuals

The plot showing residuals versus time exhibits a wave-like pattern, where there are alternating larger and smaller oscillations. This suggests a periodic structure in the residuals, but without a clear, consistent maximum or minimum. The presence of smaller variations within the larger waves indicates that there might be multiple frequencies influencing the residuals, which could suggest some form of autocorrelation.

The plot of residuals versus their lagged values suggests a somewhat linear relationship, though the data points are scattered across a wide range. The spread of points indicates that while there is some dependence between current and lagged residuals, the relationship is not perfectly linear. The absence of points in the upper left and lower right quadrants suggests

that there might be a tendency for positive autocorrelation, as most points lie in the upper right (positive residuals correlate with positive lagged residuals). This reinforces the idea that there could be first-order autocorrelation in the residuals.

10.2 Durbin-Watson Test

To formally test for the presence of first-order autocorrelation, we conduct the Durbin-Watson (DW) test, designed to detect autocorrelation in the residuals of a regression model. The test statistic ranges from 0 to 4, where:

- A DW statistic of 2 indicates no autocorrelation.
- A value less than 2 indicates positive autocorrelation.
- A value greater than 2 indicates negative autocorrelation.

$$DW = \frac{\sum_{t=2}^T (e_t - e_{t-1})^2}{\sum_{t=1}^T e_t^2}$$

where e_t represents the residual at time t , and T is the total number of observations.

From the provided data, we have the following values:

$$DW = \frac{0.00321439}{0.00789819} = 0.406978$$

Additionally, we can approximate the Durbin-Watson statistic using the formula:

$$DW_{\text{approximated}} = 2 \times (1 - \rho)$$

Where ρ is the estimated autocorrelation coefficient of the residuals. Using the provided value:

$$DW_{\text{approximated}} = 2 \times (1 - \rho) = 0.428963$$

This approximated value suggests that the level of autocorrelation in the residuals is moderate, and the Durbin-Watson statistic is consistent with the presence of positive autocorrelation.

10.3 Breusch-Godfrey Test (LM Test)

We perform the Breusch-Godfrey test for autocorrelation, which is equivalent to the LM test for higher-order autocorrelation. The steps involved in the computation are as follows:

Step 1: Estimate the OLS Model and Extract Residuals

The first step is to run the OLS regression and extract the residuals ($\hat{u}_t$):

$$\text{lgdp} = \beta_0 + \beta_1 \cdot \text{lcapital} + \beta_2 \cdot \text{laverage_labor} + \beta_3 \cdot \text{lproductivity} + \beta_4 \cdot \text{new_int} + \beta_5 \cdot \text{internet} + u$$

Step 2: Estimate the Auxiliary Equation

The second step is to estimate the auxiliary equation including the lagged residuals:

$$\hat{u}_t = \gamma_0 + \gamma_1 \cdot \text{lcapital} + \gamma_2 \cdot \text{laverage_labor} + \gamma_3 \cdot \text{lproductivity} + \gamma_4 \cdot \text{new_int} + \gamma_5 \cdot \text{internet} + \gamma_6 \cdot \hat{u}_{t-1} + v_t$$

Step 3: Compute the LM Statistic

The LM statistic for autocorrelation is calculated as:

$$LM = (T - p) \cdot R^2$$

Where:

$$T = 66, \quad p = 1, \quad R^2 = 0.644710$$

Substituting these values:

$$LM = (66 - 1) \cdot 0.644710 = 65 \cdot 0.644710 = 44.4698$$

Step 4: Find the Critical Value

We now compute the critical value from the chi-squared distribution with 1 degree of freedom and $\alpha = 0.05$:

$$\chi_{0.05}^2(1) = 3.84146$$

Step 5: Compute the p-value

Finally, we compute the p-value for the LM statistic using the chi-squared distribution:

$$p_{\chi_{LM}^2} = 2.58312 \times 10^{-11}$$

Since the p-value is much smaller than 0.05, we reject the null hypothesis of no autocorrelation and conclude that there is significant autocorrelation in the residuals.

10.4 Correction for Autocorrelation

If autocorrelation is detected, we need to correct the model to account for it. A common approach to correct for autocorrelation is Generalized Least Squares (GLS), which transforms the model to account for the correlation structure in the errors.

10.4.1 Generalized Least Squares (GLS) Estimation

The GLS method is a generalization of Ordinary Least Squares (OLS) that accounts for possible heteroscedasticity or autocorrelation in the error terms. It provides efficient estimates when the errors exhibit non-constant variance or correlation. While there are multiple GLS methods available, we chose to use the Static GLS model for our analysis.

The choice of Static GLS is driven by its ability to handle situations where the error terms are correlated over time but remain stationary. This is especially relevant when dealing with time-series data, where such correlations are common. The Static GLS assumes that the structure of the error covariance matrix does not change over time, making it an appropriate choice for our study where the focus is on analyzing the effects of productivity and related factors over a fixed period without the need to account for more complex dynamic structures.

As we already had calculated ρ before, we just have to transform our variables :

$$Y_t - \hat{\rho}Y_{t-1} = b_0 + a_1[X_{1t} - \hat{\rho}X_{1t-1}] + a_2[X_{2t} - \hat{\rho}X_{2t-1}] + v_t \quad (26)$$

10.4.2 GLS Estimation Results

Variable	Coefficient
lcapital _t	0.135065
laverage_labor _t	0.552396
lproductivity _t	0.795858
new_int _t	0.0835169
internet _t	0.0128206

Table 11: Static GLS Results

Variable	Coefficient
lcapital	0.106506
laverage_labor	0.589220
lproductivity	0.952719
new_int	0.0877519
internet	0.0161190

Table 12: OLS Results

These differences illustrate the importance of using the GLS method to account for potential issues in the error structure, leading to more efficient and potentially more accurate estimates.

We can compare these GLS results to the original OLS estimates. These results indicate that the coefficient estimates for both labor and productivity have shifted slightly compared to the OLS results. Moreover, the significance levels of both coefficients have increased, suggesting that accounting for autocorrelation has improved the precision of the estimates. In particular, the p-values for labor and productivity have decreased, implying that the variables are now more statistically significant. This demonstrates that correcting for autocorrelation has led to more accurate and robust estimates, reflecting a clearer relationship between the variables in the presence of serial correlation.

11 Detection and Correction of Heteroscedasticity

Heteroscedasticity refers to the situation where the variance of the errors in a regression model is not constant across all levels of the independent variable(s). In the context of our macroeconomic time series analysis, specifically within the Solow-Swan growth model, heteroscedasticity can arise due to various factors, such as changes in economic conditions over time, differing levels of economic growth, or structural shifts in the economy. This variability in the error terms can lead to inefficient estimates and misleading statistical inference, particularly in models dealing with time-series data.

In this section, we first detect heteroscedasticity using two methods: the White test, which checks for general forms of heteroscedasticity, and the Goldfeld-Quandt test, which compares variances in different subgroups of the data. Following the detection of heteroscedasticity, we proceed to correct it using Generalized Least Squares (GLS), a technique that adjusts the regression to account for non-constant error variance, providing more reliable coefficient estimates and improved model accuracy.

11.1 Preliminary Diagnostic: Residual Plot Analysis

The residual plot analysis serves as a crucial initial diagnostic tool for detecting potential heteroscedasticity. We plotted the residuals $\hat{\varepsilon}_i$ against the key explanatory variable `laverage_labor` to visually inspect the variance structure of the regression model.

Upon meticulous examination of the residual plot, we observe several critical characteristics:

- **Variance Dispersion:** The residuals exhibit a markedly non-uniform spread across different values of `laverage_labor`.
- **Variance Trend:** There is a clear heterogeneous variance pattern:
 1. At lower values of `laverage_labor`, residuals are tightly clustered with minimal spread.
 2. As `laverage_labor` increases, the residual variance dramatically expands, showing a fan-like or cone-like dispersion.
- **Structural Implications:** The increasing variance suggests that:
 - The model's predictive accuracy varies systematically with `laverage_labor`
 - Observations at higher `laverage_labor` values contribute disproportionately more uncertainty to the regression

Preliminary Conclusion: The residual plot strongly indicates the presence of heteroscedasticity, necessitating further formal statistical tests.



Figure 9: Errors distribution for Labor

11.2 Formal Heteroscedasticity Detection: White Test

Theoretical Foundation The White test is a general specification test for heteroscedasticity that does not assume a specific functional form of the variance.

Null Hypothesis

$$H_0 : \text{Homoscedasticity exists (constant variance of residuals)} \quad (27)$$

Test Procedure The White test involves an auxiliary regression:

$$\hat{\varepsilon}^2 = \beta_0 + \sum_{i=1}^k \beta_i x_i + \sum_{i=1}^k \beta_{i+k} x_i^2 + \varepsilon \quad (28)$$

Where:

- $\hat{\varepsilon}^2$ are squared residuals from the original regression
- x_i are explanatory variables
- k is the number of explanatory variables

Statistic	Value
R^2 from auxiliary regression	0.237
LM statistic	15.632
Degrees of freedom	2
Critical value ($\alpha = 0.05$)	5.991
p -value	0.0004

Comprehensive Interpretation

- The LM statistic (15.632) substantially exceeds the critical value (5.991)

- Extremely low p -value ($0.0004 < 0.05$) provides overwhelming evidence to reject homoscedasticity
- Significant squared terms indicate non-linear variance dependencies:
 1. lproductivity^2 ($p = 0.037$)
 2. new_int^2 ($p = 0.020$)
 3. internet ($p = 0.046$)

11.3 Formal Heteroscedasticity Detection: Goldfeld-Quandt Test

Theoretical Framework The Goldfeld-Quandt test directly compares residual variances across different subsamples of the data.

Methodology

1. Order observations by the suspected heteroscedasticity-inducing variable
2. Remove middle observations (typically 25-33%)
3. Compute separate regressions for lower and upper subsamples
4. Compare their residual sum of squares (RSS)

Test Statistic Derivation

$$F_{GQ} = \frac{SSR_2/df_2}{SSR_1/df_1} \quad (29)$$

Test Metric	Value
F_{GQ} statistic	7.952
Critical value ($\alpha = 0.05$)	3.020
p -value	0.0016
Degrees of freedom (lower sample)	10
Degrees of freedom (upper sample)	9

Detailed Results

Comprehensive Interpretation

- F_{GQ} statistic (7.952) dramatically exceeds critical value (3.020)
- Extremely low p -value ($0.0016 < 0.05$) confirms heteroscedasticity
- Substantial variance difference between lower and upper data subsamples

11.4 Heteroscedasticity Correction: Weighted Least Squares (WLS)

WLS provides an optimal estimation technique when variance is non-constant.

Parameter	OLS	WLS
laverage_labor	3.149	3.154
lproductivity	−12.672	−12.712
new_int	1.293	1.296
internet	−0.269	−0.269
R^2	0.992	0.988

Transformation Approach

$$Y_T = \frac{y}{\sqrt{x}}, \quad X_T = \frac{x}{\sqrt{x}} \quad (30)$$

- Minimal coefficient variations indicate robust estimation
- Slight differences reflect variance stabilization
- Preserved statistical significance across transformations

Key Insights

1. WLS provides more efficient estimates under heteroscedasticity
2. Transformation stabilizes variance without fundamentally altering model structure
3. Preservation of R^2 suggests consistent explanatory power

12 Conclusion

In this econometric study, we examined how internet-driven productivity growth affects labor's marginal contribution to GDP, offering key insights into the evolving link between technology, labor, and output.

Firstly, our findings underscore the transformative impact of the internet on labor's role in the economy. The structural break in the mid-1990s, driven by digital technology adoption, significantly altered labor's elasticity in the production function. This shift reflects the growing importance of productivity gains, which have amplified labor's contribution to GDP. The interaction terms in our models show that productivity improvements, enabled by the internet, have made labor more efficient and adaptable.

Moreover, our analysis of heterogeneity across different phases of internet penetration reveals that the benefits of digitalization are not uniformly distributed. The distinction between low and high internet penetration phases highlights that the full potential of digital technologies is realized only when they are deeply integrated into economic activities. This nuanced understanding is crucial for policymakers aiming to maximize the benefits of technological adoption.

Our model selection process, guided by information criteria such as AIC and SC, has led to the identification of a parsimonious yet comprehensive model that balances goodness of fit with theoretical coherence. The exclusion of the constant term in our final model aligns with the economic principle that GDP is driven solely by productive factors—capital, labor, and productivity—without any baseline output in the absence of these inputs.

The forecasting exercise for the period 2014–2019 further validates the robustness of our model. The close alignment between predicted and actual GDP values, with errors consistently below 1.5%, demonstrates the model's predictive accuracy. This highlights the importance of incorporating productivity and interaction terms in capturing the long-term trends in economic growth.

Addressing methodological challenges, such as multicollinearity and autocorrelation, has been instrumental in ensuring the reliability of our estimates. The application of ridge regression and Generalized Least Squares (GLS) techniques has mitigated these issues, providing more stable and precise coefficient estimates.

In conclusion, our study provides compelling evidence that the internet-driven productivity acceleration has reshaped labor's contribution to GDP growth. The findings underscore the importance of technological adoption and productivity enhancements in driving long-term economic expansion. As we continue to navigate the digital age, policymakers and researchers must remain cognizant of these dynamics to foster inclusive growth and harness the full potential of technological progress.

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